SC3021 Project and Laboratory Manual

College of Computing and Data Science
Nanyang Technological University

1. Objectives

The goal of the project that is supported by laboratory sessions is to have students address a real-world (or real-world like) data science problem of their choice by applying the concepts covered throughout the course. Upon completion of the assignment, students should be able to:

- Use data science thinking to address a specific data science problem.
- Design a data science solution for a specific data science problem by identifying and employing appropriate data preparation, data storage and management, modeling and analytics, and data visualization techniques.
- Implement the various components of the designed data science ecosystem using Python.
- Reflect on the developed data science application and potential issues with respect to data protection, ethics, social, and policy aspects.

2. Instructions

2.1. Team formation

The project work is **team-based**. A team consists of **two to three** students. Teams can only be formed among students of the same lab group. Teams should be formed starting in Week 1. Due to the drop period, teams may need some readjustment during Week 3. The lab supervisor may add or remove members from teams either upon request or to comply with team-size requirements. Please send him/her your requests for team readjustments. The final names of members of a student's team must be submitted together with the first deliverable.

2.2. Deliverables

Over the course of the term, students are required to submit the following three deliverables by a predefined deadline.

1. **Two days prior to the first laboratory session:** presentation **slides** (PDF, PPT) summarizing the completed data science problem formulation defined by the team, requirement analysis, and relevant data sources that have been identified. These steps are fully implemented and documented in a Python notebook, to which a link is provided in the slides. Slides should be sent to the lab supervisor.
2. **Two days prior to the second laboratory session:** presentation **slides** (PDF, PPT) summarizing the completed and justified source selection and data preparation steps. These steps are fully implemented and documented in a Python notebook, to which a link is provided in the slides. Slides should be sent to the lab supervisor.

3. **Friday of the last lecture week:** The final submission is composed of three parts, submitted in NTULearn Lab folder: (a) **Link to a Python notebook** conforming to the assignment guidelines discussed during the project presentation session, together with the **names of the team members**; (b) A short video; (c) a single PDF files for the peer assessment and AI declaration form. The implementation available on the day of the deadline will be **assessed and graded** based on the rubrics in Appendix A and the peer review for teamwork evaluation in Appendix B. Failure to submit by the deadline will result in a zero mark.

   Details on the submission are as follows:

   a) The notebook should be fully executable and cover the implementation of all data science steps, documentation, and discussion as per the **guidelines**. Thereby, it serves as both code and report submission. The link and team member names should be sent to both the lab supervisor and the course coordinator. One submission per team is sufficient. The names must match those of the students' registration.

   b) A short video (3-5 min), showing a run-through of the notebook with audio, focusing on main highlights, challenges, and insights of the project. One submission per team is sufficient. To be sent to both the lab supervisor and course coordinator.

   c) Completed and signed teamwork evaluation (Appendix B) and generative AI declaration (Appendix C). Both should be put into a single PDF. Name the file according to your student registration name. One submission per student is required. To be sent to both the lab supervisor and course coordinator.

2.3. Attendance

Attendance is **mandatory** for the two lab sessions and will be taken.

2.4. Fair participation

Each student is expected to make a **fair and equal contribution** to the project throughout the semester, and **thoroughly understand the expectations of the project**. The final marks of a team member may be adjusted based on the team score and peer review for teamwork evaluation. **Appendix B is to be submitted individually by each team member by the same deadline as Deliverable 3.**

2.5. Lab supervisors

The lab sessions are supported by a lab supervisor (usually a teaching assistant). You may approach the lab supervisor for any questions relating to the project, project submission, feedback, etc.

2.6. AI tools

NTU's policy on the use of AI tools: *"The University requires students to (i) identify any generative AI tools used and (ii) declare how the tools are used in submitted work. Please note that even with acknowledgment, copying of output generated by AI tools (in part or whole) may still be regarded as plagiarism."*

The use of generative AI (GAI) tools is permitted in this project for the purpose of assisting students in generating key ideas, generating a first text draft, refining syntax and grammar for correct language submission, and generating first code drafts. Any instance of GAI use needs to be documented by a digital paper trail in the tabular format given below. **Both Appendix C (one for each team member) and a single PDF with the digital paper trail are to be submitted together with Deliverable 3.**

GAI tool name	*e.g., Gemeni, ChatGPT, etc.*
Input prompt	*Copy the prompt you specified*
Generated output	*Screenshot of the output or link to chat transcript*
Team member	*Name of the team member responsible for this GAI use instance*
Impacted Notebook Cell	*Provide the label of the Notebook cell that was impacted by the GAI output.*

3. Introduction

The project covers the whole data science ecosystem by requiring students to implement a data science application for a **data science problem they define themselves**. The Python-based implementation covers data engineering, data analysis, and data understanding, where students implement methods and algorithms best suited for the data science question they define. Students are further required to discuss, justify, and reflect on these steps and aspects that may not be directly covered by the implementation, i.e., data protection and data ethics. Thereby, the project gives students a first **hands-on experience of solving a data science problem.**

4. Description of the project / laboratory

Students are expected to work on the project as a team outside of the classroom. Regular meetings among team members are advised to gradually discuss and implement components as the course proceeds. An exemplar guides students through what is expected an specifies guidelines to be followed. Two laboratory sessions are used for project milestone presentations and to discuss progress and next steps.

4.1. Template / exemplar

Students are provided with an exemplar Python notebook that serves as a template for the submission that is expected as part of Deliverable 3 (see Section 2.2). The content is not intended to showcase a highly novel and significant data science problem being solved by the latest and most sophisticated techniques. The main purpose is to showcase the structure and level of discussion that is expected, providing students guidance on how to present their own work.

4.2. Laboratory session 1

During the first two-hour lab session, each group will present the milestone presentation submitted as Deliverable 1. Students will receive feedback on both their team's data science problem and the considered data sources, relevant to successfully pursue the project. Presentation time, including presentation and discussion, may vary depending on the number of teams.

4.3. Laboratory session 2

During the second two-hour lab session, each group will present the milestone presentation submitted as Deliverable 2. Students will receive feedback on the data preparation and management implemented towards solving their data science problem and the resulting data quality. This is relevant to successfully pursue the project, as results of analysis highly depend on the quality of the data it is based on. Presentation time, including presentation and discussion, may vary depending on the number of teams.

A. Rubrics

Students will submit a Python Notebook comprising both code and text, forming the final report of the project. This is a team exercise.

Criteria	**Fail standard (0-40%)**	**Pass standard (41-74%)**	**High standard (75-100%)**
Identify the core definition of the problem and plan the data-riven solution. (LO 1, 2, 5)	Identifying completely wrong definitions of the problems, and planning solutions that are somewhat related but are not the actual solutions expected for the problems.	Identifying the correct and relevant definitions of the problems in line with the course materials, planning solutions reasonably in line with solutions expected for the problems, and trying to relate the course materials to the planned solutions. Accuracy and clarity can be further improved.	Identifying the correct and relevant definitions of the problems in line with the course materials, planning technically accurate steps for the solutions that are expected for the problems, and clearly connecting the course materials to the planned solutions.
Effectively process the data and devise required models to solve the problems. (LO 1, 2, 5)	Ad hoc analysis of the data and arbitrary steps processing the data and building the model without properly connecting the concepts with relevant concepts from the course. No or little evidence of critical evaluation of the proposed solution.	Logical processing of the data that demonstrates a good understanding of the concepts from the course, and building models with reasonable quality to solve the problems. Reasonable evidence of critical thinking related to the proposed solution, and producing solutions with some degree of intuition and justification for most components.	Clear logical flow of data processing that demonstrates a good understanding of the concepts from the course (and beyond), and building high-quality models to solve the problems. Extensive evidence of critical thinking related to the proposed solution, and producing solutions with clear intuition and proper justification for all components.
Overall Editorial Standard of the Solution and the Final Report. (LO 5)	Disorganized format and arrangement of the code and report, without any comment or little/no mention of references/resources.	Clear logical flow and well-formatted arrangement of the code and report, with all essential components. Reasonable comments and reasonable documentation of references /resources.	Clear logical flow and well-formatted arrangement of the code and report, with all essential components. Detailed set of technical comments to illustrate the choices made towards the solution, and to highlight the inferences. Proper documentation of references /resources.

B. Peer assessment form

Peer Review Template for teamwork evaluation

Criteria (Weights)	**Score from 1 to 9*** (**1**: Never; **3**: Rarely; **5**: Occasionally; **7**: Frequently; **9**: Always)		
(3-members team)	*Member A*	*Member B*	*Member C*
Member name			
a. Fulfilling one's responsibilities duly (15%)			
Behaved responsibly--such as attend meetings punctually and regularly; participate in discussion; complete assigned tasks/roles punctually.	*Score from 1 to 9*		
	Qualitative comments / reasons		
b. Fulfilling one's responsibilities effectively (25%)			
Behaved and contributed effectively--such as quality of work produced; creativity of ideas; extensiveness of research and thinking.	*Score from 1 to 9*		
	Qualitative comments / reasons		
c. Managing interpersonal relationships (30%)			
Listened attentively to and sought inputs from others; helped team resolve conflicts and achieved common understanding to function effectively; promoted respect for others and differences; fostered camaraderie.	*Score from 1 to 9*		
	Qualitative comments / reasons		
d. Providing support to others to achieve goal (30%)			
Behaved fairly and ethically - such as sharing responsibilities and giving credits. Exhibited group citizenship behavior - such as helping others to learn and complete their work through guidance and encouragement; standing up for others when needed.	*Score from 1 to 9*		
	Qualitative comments / reasons		

** Score of 1 should be given only when a team member does not really deserve to be awarded any mark for the team assignment (i.e., zero mark) because the member either has not or has barely participated and/or contributed to the team assignment in any meaningful manner.*

C. Declaration on Use of Generative Artificial Intellingence (GAI)

I _____ (student name), _____@e.ntu.edu.sg (NTU email) honestly and sincerely make the following declaration in relation to the following course submission:

- Name of course:
- Course Code:
- Instructor:
- Title of Assignment/Project Submission:

In relation to the foregoing I hereby declare that, fully and properly in accordance with the project instructions I have (check all appropriate):

☐ Used GAI as permitted to assist in generating key ideas only.

☐ Used GAI as permitted to assist in generating a first text only.

☐ Used GAI as permitted to refine syntax and grammar for correct language submission only.

☐ Used GAI as permitted to assist in generating a first code draft only.

I also declare that I have :

1. Fully and honestly submitted the digital paper trail required under the project instructions; and that
2. Wherever GAI assistance has been employed in the submission in word or paraphrase or inclusion of a significant idea or fact suggested by the GAI assistant, I have acknowledged this by a footnote or code comment; and that,
3. Apart from the foregoing notices, the submission is wholly my own work.

_____ _____

Student name and signature Date